AURELIA MOTORS — TECHNICAL DESIGN DOCUMENT

End-to-End Enterprise Automotive Dealership Platform

Project:	Isocode SWE Intern Take-Home Assignment — Problem 4
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1. Executive Summary & Dealership Topology

Aurelia Motors is an enterprise-grade full-stack digital dealership platform engineered for an authorized passenger-car franchise network in Lucknow, Uttar Pradesh, India. The platform bridges public-facing digital retail (model discovery, variant comparison, transparent on-road pricing, reducing-balance financing calculators, online trade-in estimations, certified pre-owned browsing with 150-point inspection audits, and self-service test-drive and service booking) with high-efficiency internal dealership operations (role-based staff authentication, CRM lead routing, third-party portal CSV batch ingestion, response-time SLA tracking, price book governance, used car price approvals with separation of duties, and AI-assisted lead response intelligence).

1.1 Physical Dealership Topology

The dealership operates across three primary sales facilities and two integrated service workshops strategically located across the Lucknow metropolitan territory:

1. **Central Lucknow Showroom (`LKO-CENTRAL`)**: - Location: Hazratganj / Vidhan Sabha Marg, Central Lucknow, UP. - Profile: Flagship downtown retail facility handling executive, luxury EV, and high-volume commuter passenger vehicles. - Facilities: 6 new car display bays, customer delivery lounge, 4 dedicated demo fleet parking bays. 2. **Gomti Nagar Showroom (`LKO-GOMTI`)**: - Location: Vibhuti Khand, Gomti Nagar, Lucknow, UP. - Profile: Technology-centric suburban facility featuring the primary Certified Pre-Owned (Used Car) display yard and fast EV charging infrastructure. - Facilities: 8 new vehicle bays, 20+ certified used vehicle display lot, trade-in physical inspection bay. 3. **Sushant Golf City Showroom (`LKO-SGC`)**: - Location: Amar Shaheed Path / Sultanpur Road corridor, Sushant Golf City, Lucknow, UP. - Profile: High-growth suburban facility catering to modern family SUVs, crossovers, and fleet corporate clients. - Facilities: 8 display bays, demo fleet staging area. 4. **Gomti Nagar Central Workshop (`WS-GOMTI`)**: - Location: Vibhuti Khand Industrial Area, Gomti Nagar, Lucknow. - Capacity: 12 concurrent service bays, express maintenance quick-lube lane, EV battery diagnostic and diagnostic testing laboratory. 5. **Sushant Golf City Workshop (`WS-SGC`)**: - Location: Sector B, Sushant Golf City, Lucknow. - Capacity: 8 concurrent service bays, heavy overhaul and paint/body collision center.

1.2 Monorepo Architecture & Technology Choices

The project is organized as an enterprise npm workspaces monorepo enforcing modular separation of concerns: - `apps/web`: Single-Page Application (SPA) built with React 19, TypeScript, Vite 8, Tailwind CSS 4, React Router 7, and TanStack Query 5. - `apps/api`: RESTful backend service built on Node.js 22, Express 5, Prisma ORM 6, PostgreSQL 18.1, Argon2id, and Zod validation. - `packages/shared`: Shared TypeScript contracts, DTO types, financial calculation arithmetic, and integer-paise currency formatting libraries consumed symmetrically by client and server.

2. Public Site Architecture & Vehicle Catalogue

The public portal provides an intuitive, high-performance digital showroom allowing customers to explore models, compare technical specifications, select exterior finishes, and inspect real-time physical showroom inventory stock units.

2.1 Model & Variant Domain Hierarchy

The vehicle catalogue is modeled as a normalized relational hierarchy: - `VehicleModel`: Represents the top-level product nameplate (e.g. *Vantara Lumen*, *Vantara Solis*, *Vantara Aria*, *Vantara Terra*, *Vantara Ventus*). Attributes include make, model name, URL slug, body type (`HATCHBACK`, `SEDAN`, `SUV`, `CROSSOVER`), summary, detailed editorial description, and active status flag. - `VehicleVariant`: Represents a specific powertrain, trim level, and transmission configuration under a model (e.g. *Solis Elegance Long Range AWD*). Attributes include fuel type (`PETROL`, `DIESEL`, `ELECTRIC`), transmission (`MANUAL`, `AUTOMATIC`), engine displacement (cc), maximum power (bhp), peak torque (Nm), seating capacity, sort order, and URL slug. - `VehicleColour`: Represents manufacturer factory color options. Stores finish (`METALLIC`, `PEARL`, `SOLID`, `MATTE`), hex code, and showroom sort order. - `VehicleSpecification`: Structured technical attributes partitioned into six standardized domain groups: `ENGINE`, `PERFORMANCE`, `DIMENSIONS`, `SAFETY`, `COMFORT`, and `TECHNOLOGY`. The seed database provides over 280 verified technical specifications across the vehicle fleet. - `VehicleImage`: High-resolution responsive imagery with semantic alt text, view angles (exterior front three-quarter, rear, cockpit, interior), and primary display flags.

2.2 Physical Showroom Inventory Tracking

Unlike basic web brochures that display abstract vehicle models, Aurelia Motors connects the online catalogue directly to physical showroom floor inventory: - `NewVehicleInventory`: Models individual physical vehicles in stock across showrooms. Each record is tracked by a unique physical stock number (`stockNumber`, e.g. `STK-LUM-2026-0042`), assigned `showroomId`, associated `variantId`, specific exterior `colourId`, manufacturing year, and physical odometer reading. - Stock Status: Managed via the `StockStatus` enumeration (`AVAILABLE`, `RESERVED`, `SOLD`, `IN_TRANSIT`). When a vehicle is booked or invoiced, its stock status is updated in real time, preventing double-allocation and giving online shoppers authentic availability indicators per showroom location.

3. Transparent Itemized On-Road Pricing Engine

A critical pain point in automotive retail is opaque on-road price estimates. Dealerships frequently advertise ex-showroom base prices while obscuring statutory taxes, registration levies, and mandatory insurance costs. Aurelia Motors enforces transparent, itemized on-road pricing backed by versioned price books.

3.1 Strict Integer Paise Currency Standard

To eliminate floating-point precision errors (such as `0.1 + 0.2 = 0.30000000000000004`), every monetary value across the entire application—database columns, TypeScript contracts, internal calculations, and API JSON payloads—is strictly stored and manipulated as an integer number of **paise** (1 INR = 100 paise). Conversion to rupee currency strings occurs strictly at the presentation layer using the Indian digit grouping formatter (`en-IN`, e.g. ■`12,34,500`).

3.2 Price Book Versioning & Date Overlap Governance

On-road pricing is governed by the `PriceBook` and `PriceEntry` data models: - `PriceBook`: Contains a unique code, descriptive name, optional `showroomId` (allowing location-specific tariffs or dealership-wide defaults when `showroomId` is null), `effectiveFrom` timestamp, optional `effectiveTo` timestamp, and status (`DRAFT`, `PUBLISHED`, `ARCHIVED`). - Overlap Prevention: When an administrator attempts to publish a price book (`POST /api/admin/cms/price-books/:id/publish`), the server executes an overlapping range check: `sql SELECT id FROM price_books WHERE id != :id AND showroom_id IS NOT DISTINCT FROM :showroomId AND status = 'PUBLISHED' AND effective_from <= :effectiveTo AND (effective_to IS NULL OR effective_to >= :effectiveFrom); ` If an overlap exists, publication is rejected with HTTP `409 Conflict` (`DATE_OVERLAP`), ensuring no variant can ever have ambiguous tariffs on any given date. - Fallback Hierarchy: When a customer requests on-road pricing for a variant at a showroom, the system queries the showroom-specific published price book first. If no showroom-specific book covers the variant, it transparently falls back to the dealership-wide published price book (`showroomId: null`) before returning `PRICING_NOT_FOUND`.

3.3 Price Component Breakdown

Each `PriceEntry` includes an `exShowroomPaise` base amount and an itemized breakdown of `PriceComponent` rows categorized as: - `ROAD_TAX`: Computed as a basis-point percentage of ex-showroom (e.g. 800 basis points = 8.00% for Uttar Pradesh State Motor Vehicle Tax). - `INSURANCE`: 1-Year Comprehensive + 3-Year Third-Party Mandatory Motor Insurance. - `REGISTRATION`: UP RTO registration, High-Security Registration Plates (HSRP), and municipal road safety cess. - `ACCESSORIES`: Basic essential protection pack (all-weather floor mats, mud flaps, boot liner). - `OTHER`: FASTag electronic toll RFID tag issuance and extended 5-year manufacturer warranty.

Mathematical Invariants Enforced: *ext{Mandatory Total} = \sum ext{Mandatory Components} ext{Standard On-Road Price} = ext{Ex-Showroom} + ext{Mandatory Total} ext{Total With Optional} = ext{Standard On-Road Price} + \sum ext{Optional Components}*

4. Certified Pre-Owned Vehicle Architecture

The Certified Pre-Owned (Used Car) module establishes trust through rigorous inspection transparency, digital condition audit files, and data privacy protection.

4.1 150-Point Technical Inspection Framework

Every pre-owned vehicle listed on Aurelia Motors must undergo a comprehensive 150-point technical evaluation conducted by an authorized dealership technical auditor. - `UsedVehicle`: Captures make, model, variant, registration year, registration month, odometer mileage in kilometers, fuel type, transmission, ownership count (e.g. single owner), exterior color shade, price in integer paise, certified status flag, and highlighted features. - `UsedVehicleInspection`: Represents the digital inspection certificate (`reportReference`, e.g. `AUR-CERT-2026-0042`). Records the date of evaluation, vehicle overall rating (`PASS`, `ATTENTION`, `FAIL`), evaluating technical inspector name, and inspector certification ID. - `InspectionItem`: Granular checklist observations grouped into 6 standard automotive audit sections: 1. *Powertrain & Mechanical*: Engine compression, turbocharger play, transmission shifting smoothness, clutch plate wear, fluid leaks. 2. *Electrical & Electronics*: ECU diagnostic scan, battery state of health (SoH), alternator output, sensor telemetry, infotainment. 3. *Chassis & Suspension*: Strut dampening, steering rack play, ball joints, wheel alignment, subframe integrity. 4. *Braking System*: Disc rotor thickness, brake pad life remaining (mm), ABS modulator cycle, brake fluid boiling point. 5. *Body & Frame Structure*: Digital paint depth gauge readings (microns), apron/pillar factory spot-welds, absence of accidental flood/chassis damage. 6. *Interior & Cabin*: Airbag module readiness, seat upholstery condition, air conditioning delta temperature, power window regulators.

4.2 Customer Data Privacy & Masked Registration Plates

To comply with Indian data privacy standards and prevent vehicle vehicle cloning or unauthorized RTO registry scraping by aggregators, registration numbers are strictly masked in public API payloads and customer-facing interfaces using bullet characters (e.g. `UP-32 •• 1184`). Unmasked vehicle registration numbers are accessible only to authenticated staff users possessing legitimate CRM operational requirements.

5. Test-Drive Booking Engine & Concurrency Control

The test-drive booking system guarantees that a customer appointment is backed by a physical demonstrator vehicle and an available consultant slot.

5.1 Demo Fleet Allocation & Availability Matrix

- `DemoVehicle`: Physical vehicles dedicated exclusively to demonstrator service at a designated showroom. - Operating Hours: Showrooms operate test-drives daily from 10:00 AM to 6:30 PM IST in 45-minute driving slots. - Real-Time Availability: When the customer selects a showroom, vehicle variant, and date (`GET /api/test-drives/availability`), the server queries the total active demo vehicles matching that variant at that showroom, subtracts confirmed appointments overlapping each time slot, and returns exact slot-by-slot availability and remaining fleet count.

5.2 Concurrency Protection & Row-Level Locking

To prevent race conditions where two customers attempt to book the final remaining demonstrator vehicle simultaneously, the booking creation service (`POST /api/test-drives`) executes within a serialized PostgreSQL transaction using explicit row locking: `typescript await prisma.$transaction(async (tx) => { // 1. Lock demo vehicle candidates for this showroom and variant const demoVehicles = await tx.demoVehicle.findMany({ where: { showroomId, variantId, isActive: true }, select: { id: true, registrationNumber: true }, });

// 2. Count active overlapping bookings const overlapping = await tx.testDriveAppointment.findMany({ where: { showroomId, variantId, status: { in: ['CONFIRMED', 'PENDING'] }, startTime: { lt: slotEnd }, endTime: { gt: slotStart }, }, select: { demoVehicleId: true }, });

const bookedDemoIds = new Set(overlapping.map(b => b.demoVehicleId)); const availableDemo = demoVehicles.find(d => !bookedDemoIds.has(d.id));

if (!availableDemo) { throw new AppError(409, 'SLOT_UNAVAILABLE', 'Selected test-drive slot is no longer available'); }

// 3. Atomically insert confirmed appointment with allocated demo vehicle ID return tx.testDriveAppointment.create({ ... }); }); ` If two concurrent requests target the same slot, the second transaction fails the availability check and receives HTTP `409 Conflict`, guaranteeing zero double-bookings.

5.3 Idempotency, Tokens & Calendar Synchronization

- **Idempotency**: Clients supply a unique `idempotencyKey` (UUIDv4). Re-submitting the same key returns the existing booking record with HTTP 200/201, protecting against network disconnects or repeated button clicks. - **Confirmation Token**: Each booking receives a cryptographically secure random 32-character `confirmationToken`. Public booking verification (`GET /api/bookings/confirmation/:reference`) masks customer PII unless the valid token is supplied as a query parameter. - **Cancellation**: Customers can cancel their booking (`POST /api/bookings/:reference/cancel`) using the reference and token, immediately releasing the demo vehicle back to the available pool. - **Calendar Integration**: Confirmed bookings generate downloadable RFC 5545 compliant `.ics` calendar invitation files with reminder alarms and exact showroom map directions.

6. Workshop Service Booking Engine & Capacity Tracking

Automotive service operations require strict daily capacity enforcement based on physical workshop lift bays and technician operating hours.

6.1 Workshop Operations & Service Packages

- `Workshop`: Physical maintenance facility linked to operating hours and maximum concurrent bay capacity (`maxConcurrentBays`, e.g. 12 bays at Gomti Nagar, 8 bays at Sushant Golf City). - `ServiceType`: Standardized service offerings with standard labor hours, scope checklists, and base price guidelines: 1. *Express Care Service* (1.5 hours): Engine oil & oil filter replacement, 25-point visual safety inspection, fluid top-ups, windshield washer wash. 2. *Scheduled Periodic Maintenance* (3.0 hours): Complete synthetic oil change, air filter, pollen filter, spark plugs / fuel filter, 60-point mechanical check. 3. *Major Comprehensive Overhaul* (5.0 hours): Transmission fluid flush, coolant exchange, brake pad & disc rotor renewal, suspension bushing inspection. 4. *Brake & Suspension Health Service* (2.5 hours): Full hydraulic brake bleed, caliper pin lubrication, shock absorber dyno test. 5. *EV Battery & High-Voltage Health Audit* (2.0 hours): High-voltage cell impedance analysis, thermal cooling loop pressure test, charging port diagnostics.

6.2 Bay Capacity Limits & Overflow Prevention

When booking a service appointment (`POST /api/services/bookings`), the server aggregates existing active appointments scheduled for the workshop during the target shift: `typescript const concurrentBookings = await tx.serviceAppointment.count({ where: { workshopId, status: { in: ['CONFIRMED', 'IN_PROGRESS'] }, startTime: { lt: serviceEndTime }, endTime: { gt: serviceStartTime }, }, });

if (concurrentBookings >= workshop.maxConcurrentBays) { throw new AppError(409, 'WORKSHOP_CAPACITY_EXCEEDED', 'All service bays are fully booked for this time window'); } ` If bay capacity is

exceeded, the API cleanly rejects the request with HTTP `409 Conflict` (`WORKSHOP_CAPACITY_EXCEEDED`), preventing workshop over-scheduling.

7. Financial Calculations & Online Valuation Tools

To empower customers during purchase consideration, Aurelia Motors integrates an interactive reducing-balance EMI calculator and an online vehicle trade-in valuation tool.

7.1 Reducing-Balance EMI Calculation Formula

Automotive retail loans in India use the reducing-balance monthly amortization formula. The standard monthly installment E is calculated as: $E = P \cdot r \cdot \frac{(1+r)^n}{(1+r)^n - 1}$ Where: - P = Principal loan amount in paise ($P = \text{On-Road Price in paise} - \text{Down Payment in paise}$) - r = Monthly interest rate ($\frac{\text{Annual Interest Rate in Basis Points}}{100 \times 100 \times 12}$) - n = Loan duration in months (e.g. 12 to 84 months)

Amortization Properties: $\text{Total Repayment Paise} = E \times n$ $\text{Total Interest Payable Paise} = \text{Total Repayment Paise} - P$ All calculations in `@aurelia/shared/finance.ts` use pure integer paise arithmetic with zero floating-point accumulation errors. The UI features interactive sliders for down payment, interest rate, and loan tenure with instantaneous amortization graph updates.

7.2 Vehicle Trade-In Valuation Engine

Customers can receive an instantaneous valuation band for their existing vehicle without requiring user account creation: - Input parameters: Make, model, variant, registration year, odometer mileage, physical condition (`EXCELLENT`, `GOOD`, `FAIR`, `POOR`), and ownership count. - Depreciation Model: Applies age-based exponential depreciation curves calibrated against real Indian used vehicle market benchmarks: $\text{Depreciation Rate} = \text{Base Year Factor} \times (1 + \text{Mileage Penalty}) \times \text{Condition Multiplier}$ - Valuation Band: Returns a realistic indicative spread (`estimatedValuationMinPaise` and `estimatedValuationMaxPaise`) along with an explicit disclaimer that formal valuations require showroom physical inspection. - Trade-In Request Submission: Customers submit formal trade-in requests with DPDP privacy consent (`POST /api/trade-ins`), generating a tracking reference (`TRD-xxxx`) and automatically feeding into the dealership CRM pipeline.

8. Staff Authentication, Session Security & RBAC

The internal dealership management portal (`/staff`) is protected by multi-layered enterprise authentication and strict role-based access control (RBAC).

8.1 Password Hashing with Argon2id

All staff passwords are authenticated using **Argon2id** (via native `argon2` bindings), the gold-standard memory-hard hashing algorithm resistant to GPU-based brute-force and side-channel timing attacks. Passwords must satisfy strict complexity requirements (minimum 8 characters, uppercase, lowercase, number, and special character).

8.2 Stateful HttpOnly Sessions & Cryptographic CSRF

- **Session Cookie**: Staff authentication sets an `aurelia_session` cookie configured with: - `httpOnly: true` (prevents JavaScript access and XSS cookie exfiltration) - `sameSite: 'lax'` (guards against cross-site request forgery) - `secure: true` in production environments (enforces HTTPS transmission) - **Token Storage**: The raw session token is never stored in plain text. Only its SHA-256 cryptographic digest is stored in the `staff_sessions` database table, ensuring a database compromise cannot reveal active session tokens. - **Double-Submit HMAC-SHA256 CSRF**: Mutating endpoints (`POST`, `PUT`, `PATCH`, `DELETE`) verify a double-submit `X-CSRF-Token` header signed with a server secret (`CSRF_SECRET`). Omitting or tampering with the token results in immediate rejection with HTTP `403 Forbidden` (`CSRF_ERROR`).

8.3 Role-Based Access Control (RBAC) & Facility Boundaries

The platform defines 5 distinct staff roles: 1. `ADMIN`: Full dealership-wide administrative rights across all 3 showrooms and 2 workshops. Can manage staff directory, configure price books, approve used-car prices, view global metrics, and perform CSV imports. 2. `SALES_CONSULTANT`: Scoped to an assigned showroom location (e.g. Central Lucknow or Gomti Nagar). Can access leads, test-drives, and inventory belonging strictly to their assigned facility. 3. `SERVICE_ADVISOR`: Scoped to an assigned service workshop (e.g. Gomti Workshop). Can view, update, and manage workshop service appointments and bay allocations. 4. `USED_CAR_SPECIALIST`: Manages certified pre-owned vehicle stock, inspects trade-ins, and submits used car price proposals. 5. `REVIEWER`: Read-only audit access for evaluation and compliance verification.

8.4 Last Administrator Protection Safeguard

To prevent catastrophic accidental lockout, the staff administration service enforces an invariant: the last remaining active administrator account in the dealership cannot be disabled, demoted to another role, or deleted. Any attempt returns HTTP `400 Bad Request` (`CANNOT_DEMOTE_LAST_ADMIN`).

9. Omnichannel Customer Enquiry Pipeline & DPDP Act 2023 Compliance

The Digital Personal Data Protection Act, 2023 (DPDP Act) governs all collection and processing of personal data in India. Aurelia Motors implements privacy-by-design principles across all customer touchpoints.

9.1 Mandatory Affirmative Consent & Audit Trail

Every public form capturing personal information (Vehicle Enquiries, Test-Drive Bookings, Service Bookings, Trade-In Requests) strictly enforces explicit, affirmative consent: - Non-Preticked Consent: The consent checkbox is unchecked by default. The Zod validation schema strictly requires `privacyConsent: true` (or `consentGranted: true`). Submitting `false` or omitting consent results in HTTP `422 Unprocessable Entity`. - Legal Notice: Clear, prominent legal disclosure informs the data principal that their data will be processed strictly for consultation, test-drive coordination, or servicing under Aurelia Motors Lucknow authorized facilities. - Consent Audit Logging: The system creates an immutable `CustomerConsent` record storing: - `customerId`: Foreign key to customer record. - `purpose`: Enumerated processing purpose (`CUSTOMER_ENQUIRY`, `TEST_DRIVE_BOOKING`, `SERVICE_BOOKING`, `TRADE_IN_EVALUATION`). - `ipHash`: SHA-256 salted hash of the customer's remote IP address. - `userAgent`: Client browser User-Agent header. - `consentedAt`: ISO-8601 UTC timestamp.

9.2 Rate Limiting & Anti-Spam Throttling

Public inbound endpoints are protected against abuse: - Inbound Enquiry Rate Limiter: Enforces a sliding-window ceiling of 5 requests per 15-minute window per IP address. Exceeding the threshold returns HTTP `429 Too Many Requests`. - Duplicate Enquiry Prevention: If a customer submits an identical enquiry for the same showroom within 30 minutes, the submission is rejected with HTTP `409 Conflict` (`DUPLICATE_ENQUIRY`), preventing accidental multi-click duplicates while guiding the customer to await consultant follow-up.

10. Deterministic Workload-Balanced Lead Routing Engine

Lead distribution speed directly impacts automotive conversion rates. The Aurelia Motors routing engine eliminates manual assignment delays by automatically routing incoming leads within milliseconds.

10.1 Active Workload Scoring Algorithm

When an inbound customer enquiry arrives, the `LeadAssignmentService` dynamically scores all eligible sales consultants assigned to the designated showroom: *ext{Active Workload} = \sum ext{Leads in status } \{ ext{NEW}, ext{ASSIGNED}, ext{CONTACTED}, ext{QUALIFIED}, ext{TEST_DRIVE_SCHEDULED}, ext{NEGOTIATING}\}* Consultants with closed leads (`WON` or `LOST`) have those records excluded from active workload scoring.

Assignment Decision Logic: 1. Find all active `StaffUser` records with role `SALES_CONSULTANT` whose `StaffLocationAssignment` includes the enquiry's `showroomId`. 2. Compute the active workload count for each

candidate consultant. 3. Select the consultant with the minimum active workload. If multiple consultants share the lowest workload, tie-breaking is resolved deterministically by selecting the consultant whose most recent assignment was earliest (least recently assigned). 4. If no showroom-specific consultant is available, the lead is safely routed to the dealership General Administrator as an unhandled fallback.

10.2 Sub-500ms Lead Routing SLA

The lead creation and assignment pipeline executes in a single database transaction, achieving an average latency under 50 milliseconds (benchmarked at 32ms in automated testing), comfortably beating the platform's 25-second mobile 4G latency ceiling.

11. Third-Party Portal CSV Batch Import & Deduplication

Automotive dealerships receive large volumes of leads from aggregator portals (CarDekho, CarWale, Gaadi.com). Aurelia Motors provides an administrative batch ingestion interface with intelligent header matching and duplicate filtering.

11.1 Robust Header Mapping & Normalization

The CSV parser (`LeadImportService.parseCsvRows`) handles quoted values, escaped characters, and Windows (`\r\n`) / Unix (`\n`) newline discrepancies. It normalizes column headers against common portal variations: - Customer Name: `customername`, `name`, `fullname`, `customer`. - Phone: `phone`, `mobile`, `contact`, `phonenumber`. - Email: `email`, `emailaddress`. - Showroom: `showroom`, `location`, `dealership`, `facility`. - Model: `model`, `vehicle`, `car`, `interestedmodel`. - External Portal ID: `externalid`, `leadid`, `id`, `portalid`.

11.2 Two-Phase Import Architecture

1. **Preview Phase (`POST /api/staff/leads/import/preview`)**: - Parses the uploaded CSV file without database modification. - Validates mandatory fields (10-digit Indian phone number starting with 6-9, valid email syntax, customer name >= 2 characters). - Resolves showroom names against active Lucknow facilities (e.g. matching `LKO-CENTRAL` or strings containing `Central`). - Flags intra-file duplicate rows (duplicate phone + model or duplicate external ID within the file). - Checks against existing database leads: flags leads with matching `externalLeadId` or identical customer phone/email submitted within the last 7 days. - Returns row-by-row validation feedback (`validRows`, `duplicateRows`, `errorRows`). 2. **Execution Phase (`POST /api/staff/leads/import`)**: - Accepts validated rows and inserts them within an atomic database batch transaction. - Records an `ImportBatch` audit entry tracking batch reference (`IMP-xxxx`), source filename, total row count, imported count, and executing administrator. - Automatically executes the deterministic workload-balanced routing engine for each imported lead, instantly assigning them to on-duty sales consultants.

12. Lead CRM Lifecycle & Response-Time Tracking

Automotive sales excellence requires rapid human outreach. Aurelia Motors implements strict response-time tracking to maintain operational discipline.

12.1 The 30-Minute First-Human-Response SLA

Dealership policy mandates that every prospective customer receive direct human contact within **30 minutes** of submitting an enquiry (`LEAD_RESPONSE_SLA_MINUTES = 30`). - **Critical Architectural Distinction**: Automated system routing and lead creation **never** count as a first-response. The lead's `firstResponseAt` timestamp remains strictly `NULL` until a human staff member records an explicit outreach event. - **Overdue Detection**: If a lead remains without recorded contact 30 minutes after creation, the CRM automatically flags the lead as overdue (`isOverdue: true`). Overdue leads display prominent amber/red pulsing badges in the staff inbox and escalate on the dealership dashboard.

12.2 Contact Activity Logging & Audit Trail

Sales consultants record customer interactions via `POST /api/staff/leads/:id/activities`: - Activity Channels: Phone Call (`PHONE`), WhatsApp Message (`WHATSAPP`), Email (`EMAIL`), In-Person Showroom Meeting (`MEETING`). - Automated SLA Stamping: The first logged human outreach activity permanently records `firstResponseAt` and computes the exact response time in minutes. Subsequent activities are appended to the lead's historical audit timeline without modifying the initial response timestamp. - Lead Status Lifecycle: Managed via audited transitions: *ext{NEW} \longrightarrow ext{ASSIGNED} \longrightarrow ext{CONTACTED} \longrightarrow ext{QUALIFIED} \longrightarrow ext{TEST_DRIVE_SCHEDULED} \longrightarrow ext{NEGOTIATING} \longrightarrow ext{WON} ext{ or } ext{LOST}* Every status change appends a row to `lead_status_history` recording the prior status, new status, timestamp, and acting staff user ID.

13. Dealership Content & Price Governance CMS

The platform empowers dealership administrators to manage marketing campaigns, tariffs, and inventory without engineering intervention.

13.1 Administrative CMS Capabilities

- Vehicle Model & Variant CMS (`/api/admin/cms/models`, `/api/admin/cms/variants`): Administrators can create new models, configure powertrain specifications, upload imagery, and soft-deactivate discontinued trims. - Showroom Inventory Management (`/api/admin/cms/inventory`): Add, inspect, and update physical stock units assigned to showrooms. - Certified Used Vehicle CMS (`/api/admin/cms/used-vehicles`): Provision certified pre-owned stock, manage condition findings, and adjust list prices. - Banners & Promotional Offers (`/api/admin/content/banners`, `/api/admin/content/offers`): Configure homepage marketing carousels, festival finance schemes (e.g. *Festive Season Zero Down-Payment Drive*), exchange bonuses, and interest rate subsidies.

13.2 Used Car Price Approval Governance & Separation of Duties

To protect dealership gross margins and prevent unauthorized price tampering, Aurelia Motors enforces a strict separation-of-duties workflow for certified pre-owned pricing: 1. **Proposal Submission**: A used car specialist submits a price adjustment proposal (`POST /api/staff/used-vehicles/:id/propose-price`), specifying the target price in integer paise and business justification (e.g. *Market competitive adjustment following 30-day inventory aging*). 2. **Review & Approval**: The proposed price is held in `PENDING` state. Only a Dealership Administrator can approve or reject the proposal (`POST /api/staff/used-car-pricing/proposals/:id/approve`). 3. **Drift Protection**: If approved, the vehicle's `pricePaise` is atomically updated, an audit record is sealed, and real-time notifications are dispatched to staff.

14. AI-Assisted Lead Operations Architecture

Aurelia Motors integrates intelligent lead assistance while maintaining strict architectural honesty and data integrity.

14.1 Honest Architectural Labelling: Grounded Domain Engine vs External LLM

Many systems misleadingly label hardcoded templates as "generative AI" or deploy ungrounded external LLMs that hallucinate non-existent car models and false prices. Aurelia Motors takes an honest, robust approach: - **Default Architecture**: Powered by the **Aurelia Grounded Domain Engine**, an in-process deterministic intelligence engine combining keyword intent extraction, urgency scoring, contextual multi-channel templating, and live PostgreSQL relational stock matching. - **Optional External LLM**: If the dealership configures an `AI_API_KEY` (Google Gemini 1.5 Flash), the system activates external neural synthesis for draft generation while strictly grounding the prompt with real database stock rows and sanitizing customer PII. - **Provider Transparency**: The API transparently reports its active intelligence provider (`provider: 'Aurelia Grounded Domain Engine (Deterministic Rule-Based & DB-Matched)'` or `'Gemini 1.5 Flash (Grounded LLM)'`).

14.2 Grounded Lead Operations Features

1. **Structured Lead Intent & Urgency Summary (**`POST /api/staff/ai/:leadId/summary`**)**: - Parses customer notes and enquiry parameters. - Evaluates urgency (`HIGH`, `MEDIUM`, `LOW`) based on customer timing indicators (e.g. wedding, immediate delivery, festival purchase). - Recommends concrete next steps for the assigned sales consultant. 2. **Human-in-the-Loop Multi-Channel Response Drafts (**`POST /api/staff/ai/:leadId/response-draft`**)**: - Generates customized drafts tailored for **WhatsApp** (casual greeting with emoji and quick-action link), **Email** (formal quotation and showroom brochure), and **Phone Call Script** (structured discovery guide with opening, probing questions, and appointment close). - Human-in-the-Loop Mandate: Drafts are never sent automatically. They populate an editable textarea in the staff CRM interface, allowing the consultant to review, refine, and approve before communicating. 3. **Real Stock-Matched Vehicle Recommendations (**`POST /api/staff/ai/:leadId/recommendations`**)**: - Queries live PostgreSQL inventory rows (`priceEntries` and `usedVehicles`). - Recommends only authentic in-stock models within the customer's budget band. - **Zero Hallucination Guarantee**: Prices, specifications, and availability are strictly retrieved from real database records and never hallucinated.

15. Notification System & Outbound Communication Boundaries

Effective dealership coordination requires prompt internal staff alerts and clear outbound customer acknowledgements.

15.1 Real-Time Staff Notifications Popover

The staff operational header features a real-time notification popover backed by the `StaffNotification` entity: - Event Triggers: - `LEAD_ASSIGNED`: Alerts a sales consultant immediately upon lead allocation. - `PRICE_PROPOSAL_SUBMITTED`: Alerts administrators of pending used car price proposals. - `PRICE_PROPOSAL_DECIDED`: Alerts specialists when their price proposal has been approved or rejected. - `TRADE_IN_EVALUATED`: Alerts consultants when formal trade-in valuation offers are issued. - API Endpoints: `GET /api/staff/notifications`, `GET /api/staff/notifications/unread-count`, `PATCH /api/staff/notifications/:id/read`, `POST /api/staff/notifications/mark-all-read`.

15.2 Outbound Communication Boundaries & SMS Webhook Architecture

- Outbound Acknowledgements: When appointments or enquiries are submitted, the system generates downloadable `.ics` calendar files, on-screen confirmations, and formatted SMS/WhatsApp message payloads. - **Production Integration Boundary**: In production enterprise deployments, outbound SMS and WhatsApp messages are dispatched via third-party telecom gateways (e.g. Twilio, Gupshup, Exotel). In this take-home implementation, outbound messaging is cleanly structured as an event webhook interface, avoiding paid third-party external dependencies while maintaining production-ready integration contracts.

16. Comprehensive Data Model & PostgreSQL Architecture

The database schema comprises over 30 normalized tables implemented in PostgreSQL 18.1 with Prisma ORM 6:

16.1 Primary Relational Entities

1. **Locations & Facilities**: `showrooms`, `workshops`, `workshop_operating_hours`. 2. **Vehicle Catalogue**: `vehicle_models`, `vehicle_variants`, `vehicle_colours`, `vehicle_images`, `vehicle_specifications`. 3. **Physical Inventory**: `new_vehicle_inventory`, `used_vehicles`, `used_vehicle_inspections`, `used_vehicle_inspection_items`, `demo_vehicles`. 4. **Tariffs & Pricing**: `price_books`, `price_entries`, `price_components`, `used_car_price_proposals`. 5. **Appointments**: `test_drive_appointments`, `service_appointments`, `service_types`. 6. **Customers & Privacy**: `customers`, `customer_consents`. 7. **Staff & Administration**: `staff_users`, `staff_sessions`, `staff_location_assignments`, `auth_audit_logs`, `staff_notifications`. 8. **CRM & Enquiries**: `leads`, `lead_activities`, `lead_notes`, `lead_status_history`, `import_batches`. 9. **Marketing Content**: `homepage_banners`, `promotion_offers`, `trade_in_requests`.

16.2 Key Database Constraints & Performance Indexes

- Composite Unique Constraints: - `@@unique([priceBookId, variantId])` on `price_entries` - `@@unique([priceEntryId, code])` on `price_components` - `@@unique([variantId, group, label])` on `vehicle_specifications` - `@@unique([modelId, name])` on `vehicle_colours` - B-Tree Performance Indexes: - `@@index([showroomId, status])` on `new_vehicle_inventory` - `@@index([registrationYear, mileageKm])` on `used_vehicles` - `@@index([showroomId, variantId, startTime, status])` on `test_drive_appointments` - `@@index([workshopId, startTime, status])` on `service_appointments` - `@@index([showroomId, assignedStaffId, status])` on `leads` - `@@index([customerId, purpose, consentedAt])` on `customer_consents`

17. Security Architecture & Threat Modeling

The application is hardened against common web application vulnerabilities (OWASP Top 10):

17.1 Vulnerability Mitigations

- **SQL Injection**: Neutralized completely through Prisma ORM's parameterized query engine and strict TypeScript type boundaries. Zero raw string interpolation queries exist in production paths. - **Cross-Site Scripting (XSS)**: React 19's virtual DOM auto-escapes rendered text. Security headers (`X-Content-Type-Options: nosniff`, `X-XSS-Protection: 1; mode=block`, `Content-Security-Policy`) prevent inline script injection. - **Cross-Site Request Forgery (CSRF)**: Double-submit HMAC-SHA256 tokens verified on all state-changing endpoints. - **Insecure Direct Object References (IDOR)**: Staff endpoints verify both authentication and location scoping. A sales consultant assigned to Gomti Nagar cannot query or mutate leads belonging to Central Lucknow (`403 Forbidden`). - **Brute-Force & Denial of Service**: Rate limiting applied to login endpoints (5 attempts / 15 minutes) and enquiry endpoints. Session fixation is prevented by regenerating session tokens upon login.

18. Performance Verification & Scalability

Performance is verified under simulated production conditions: - **Public Page Load Latency**: Server-side response times for catalogue models and on-road pricing average under 25 milliseconds. Production web assets are compiled into optimized, code-split chunks with Gzip/Brotli compression, achieving sub-2-second Time to Interactive (TTI) on standard 4G mobile connections. - **Lead Assignment Speed**: Inbound lead routing executes in under 50 milliseconds (benchmarked at 32ms in automated tests), comfortably within the platform's 25-second mobile 4G latency ceiling. - **Database Connection Pooling**: Prisma Client manages a persistent connection pool configured with timeout parameters, health probes, and automatic reconnection logic.

19. Responsive UI Design & Accessibility (a11y)

The public site and staff portal are responsive and accessible across device viewports: - **Viewport Support**: - *Mobile (375px)*: Compact stacked navigation, sticky booking CTAs, full-width modal sheets, touch-friendly tap targets (minimum 44x44px). - *Tablet (768px)*: 2-column vehicle discovery grids, side-by-side pricing breakdowns. - *Desktop (1440px)*: High-density data tables, multi-column CRM kanban workflows, split-screen inspection report viewers. - **Accessibility (WCAG 2.1 AA)**: - Semantic HTML landmarks (`<header>`, `<nav>`, `<main>`, `<article>`, `<section>`, `<footer>`). - High-contrast typography pairing editorial serif headings (*Fraunces*) with geometric sans-serif UI type (*Manrope*). - Keyboard Navigation: All buttons, dropdowns, and modal dialogs are fully keyboard-navigable with visible focus rings (`focus-visible:ring-2`). - Screen Reader Support: Form inputs include explicit `<label>` bindings and descriptive ARIA roles.

20. Operational Runbook & Reviewer Evaluation Guide

Reviewers can evaluate the application using Docker Compose or native local execution:

20.1 One-Click Docker Compose Evaluation

`bash docker compose up --build ` - Public Site: `http://localhost:5173` or `http://localhost` - REST API: `http://localhost:4000/api` - API Health Status: `http://localhost:4000/api/health` - Staff Portal Login: `http://localhost:5173/staff/login`

20.2 Native Local Evaluation

`bash # 1. Install dependencies npm install

2. Configure environment cp .env.example .env

3. Apply database migrations & seed initial dataset npm run db:deploy npm run db:seed

4. Start local development servers npm run dev `

20.3 Pre-Seeded Evaluation Accounts

All accounts use the demo password `Aurelia2026!`: - Administrator: `admin@aureliamotors.com` (Global dealership access) - Sales Consultant (Central): `sales.central@aureliamotors.com` (Scoped to Central Lucknow) - Sales Consultant (Gomti Nagar): `sales.gomti@aureliamotors.com` (Scoped to Gomti Nagar) - Service Advisor (Gomti Workshop): `service.gomti@aureliamotors.com` (Scoped to Gomti Workshop) - Service Advisor (Sushant Golf City Workshop): `service.sgc@aureliamotors.com` (Scoped to SGC Workshop) - Used Car Specialist: `usedcars@aureliamotors.com` (Certified Pre-Owned yard)

Note: The staff login screen includes 1-click quick-fill buttons for instant reviewer credential population.

21. Test Suite & Verification Summary

The codebase has undergone comprehensive automated verification: - **Automated Test Results**: **126 passing tests across 15 test suites** executed via Vitest and Supertest: 1. `validation.test.ts` (2 tests): Request validation schemas and error envelope formatting. 2. `health.test.ts` (7 tests): Database connection health probe and uptime reporting. 3. `showrooms.test.ts` (3 tests): Showroom listing, code lookup, and active facility filtering. 4. `catalogue.test.ts` (8 tests): Vehicle catalogue discovery, variant specs, and on-road pricing. 5. `used-vehicles.test.ts` (5 tests): Pre-owned listings, inspection reports, and masked registration plates. 6. `trade-ins.test.ts` (5 tests): Valuation heuristics, DPDP consent enforcement, and trade-in submission. 7. `appointments.test.ts` (17 tests): Test-drive capacity, double-booking race condition prevention, workshop bay capacity, token lookup, and cancellation. 8. `auth.test.ts` (25 tests): Staff login, Argon2id verification, session revocation, CSRF protection, and location scoping. 9. `leads.test.ts` (14 tests): Inbound enquiry intake, deterministic routing, 30-minute response-time tracking, overdue detection, and CSV batch import. 10. `cms.test.ts` (7 tests): Vehicle model & variant CMS, showroom inventory stock, price book publishing, and date overlap prevention. 11. `content.test.ts` (4 tests): Homepage banners and promotional campaign CMS. 12. `notifications.test.ts` (5 tests): Staff notifications popover, unread counts, and mark-as-read workflows. 13. `used-car-pricing.test.ts` (6 tests): Separation-of-duties price proposals and approval governance. 14. `ai.test.ts` (4 tests): Grounded lead intent summary, multi-channel response drafts, and inventory-backed stock recommendations. 15. `e2e-journeys.test.ts` (14 tests): Full end-to-end integration journeys across all customer and staff workflows. - **Code Quality**: ESLint 9 passing with 0 warnings, Prettier 3 passing with 0 formatting warnings, strict TypeScript passing with 0 compilation errors, and production build succeeded in 907ms.

22. Out-of-Scope Boundaries & Honest System Trade-offs

To maintain architectural clarity and focus on the core requirements of Problem 4, the following boundaries were deliberately established: 1. **Financial Payment Gateway**: Test-drive appointments and workshop service bookings are complimentary in accordance with authorized dealership customer care standards. Online down-payment collection (e.g. Razorpay/Stripe) is outside the scope of Problem 4. 2. **Third-Party Paid SMS/WhatsApp Gateway**: Outbound messaging is implemented as structured notification templates and webhook boundaries rather than connecting to live paid commercial telecom aggregators. 3. **AI Integration**: The platform transparently utilizes a deterministic Grounded Domain Engine

querying real PostgreSQL stock by default, with optional external Gemini LLM activation, avoiding ungrounded AI hallucinations.

23. Future Evolution & Roadmap

Key architectural extension points for post-internship production expansion: - **Omnichannel Live Chat & WhatsApp Bot**: Integrating conversational AI directly into the customer WhatsApp channel using the existing Grounded Domain Engine. - **Automated Service Bay Telemetry**: Connecting IoT OBD-II diagnostic scanners in the workshops directly to vehicle service appointment records. - **Dynamic Insurance Premium Calculator**: API integration with Indian general insurance underwriting engines (e.g. ICICI Lombard, Bajaj Allianz) for live quotes based on vehicle IDV.

24. Appendix: Complete API Reference & Database Dictionary

24.1 Complete REST API Endpoint Directory

Method	Endpoint	Access	Purpose
:---	:---	:---	:---
GET	/api/health	Public	Service health & PostgreSQL connectivity probe
GET	/api/showrooms	Public	Lists active Lucknow showroom facilities
GET	/api/showrooms/:code	Public	Individual showroom facility detail
GET	/api/catalogue/models	Public	Model catalogue with filters, search, and starting prices
GET	/api/catalogue/models/:slug	Public	Complete model detail with specifications and variants
GET	/api/catalogue/variants/:slug/pricing	Public	Itemized on-road price breakdown with statutory taxes
GET	/api/used-vehicles	Public	Certified pre-owned vehicle listings with masked plates
GET	/api/used-vehicles/:slug	Public	Pre-owned detail with 150-point inspection findings
POST	/api/trade-ins/estimate	Public	Instant trade-in valuation band without registration
POST	/api/trade-ins	Public	Formal trade-in request with DPDP consent
GET	/api/test-drives/availability	Public	Real-time demonstrator vehicle slot availability
POST	/api/test-drives	Public	Concurrency-safe, idempotent test-drive reservation
GET	/api/services/workshops	Public	Authorized workshop directory and operating schedules
GET	/api/services/types	Public	Standard maintenance service packages and rates
GET	/api/services/availability	Public	Capacity-checked workshop bay availability
POST	/api/services/bookings	Public	Concurrency-safe service appointment booking
GET	/api/bookings/confirmation/:ref	Public	Booking verification with token-based unmasking
POST	/api/bookings/:ref/cancel	Public	Self-service cancellation releasing slot capacity
POST	/api/enquiries	Public	Omnichannel customer enquiry capture with DPDP consent
GET	/api/banners	Public	Active homepage marketing banners
GET	/api/offers	Public	Active promotional financing and exchange offers
POST	/api/auth/login	Public	Staff authentication with Argon2id & session cookie
POST	/api/auth/logout	Staff	Revokes active session and clears session cookie
GET	/api/auth/me	Staff	Returns current staff profile, role, and CSRF token
POST	/api/auth/change-password	Staff	Self-service password update with Argon2id re-hashing
GET	/api/staff/dashboard	Staff	Role- and facility-scoped operational metrics
GET	/api/staff/leads	Staff	Scoped CRM lead inbox with SLA urgency sorting
GET	/api/staff/leads/:id	Staff	Lead record, customer notes, and activity timeline
PATCH	/api/staff/leads/:id/status	Staff	Lead lifecycle status progression with audit trail
POST	/api/staff/leads/:id/activities	Staff	Outreach logger establishing first-human-response SLA
POST	/api/staff/leads/:id/notes	Staff	Internal private consultation notes
POST	/api/staff/leads/import/preview	Admin	Validates portal CSV rows and flags duplicates
POST	/api/staff/leads/import	Admin	Executes batch CSV import and automated routing
GET	/api/staff/trade-ins	Staff	Queue of trade-in vehicle requests
PATCH	/api/staff/trade-ins/:id	Staff	Issues formal trade-in valuation counter-offer
POST	/api/staff/used-vehicles/:id/propose-price	Staff	Specialist submits price adjustment proposal
GET	/api/staff/used-car-pricing/proposals	Staff	Queue of pending used car price proposals

POST	/api/staff/used-car-pricing/proposals/:id/approve	Admin	Admin approves proposal (separation of duties)
POST	/api/staff/used-car-pricing/proposals/:id/reject	Admin	Admin rejects proposal with mandatory review note
POST	/api/staff/ai/:id/summary	Staff	Grounded AI lead urgency & intent summary
POST	/api/staff/ai/:id/response-draft	Staff	Multi-channel response draft (WhatsApp, Email, Phone)
POST	/api/staff/ai/:id/recommendations	Staff	Grounded real PostgreSQL stock recommendations
GET	/api/staff/notifications	Staff	Staff notifications queue
GET	/api/staff/notifications/unread-count	Staff	Unread notifications badge counter
PATCH	/api/staff/notifications/:id/read	Staff	Marks individual notification as read
POST	/api/staff/notifications/mark-all-read	Staff	Marks all notifications as read
GET	/api/admin/cms/models	Admin	Administrative vehicle models listing
POST	/api/admin/cms/models	Admin	Creates new vehicle model
PATCH	/api/admin/cms/models/:id	Admin	Updates vehicle model attributes
POST	/api/admin/cms/models/:id/variants	Admin	Provisions new powertrain variant under model
GET	/api/admin/cms/inventory	Admin	Lists showroom physical stock units
POST	/api/admin/cms/inventory	Admin	Adds new stock unit to showroom floor
GET	/api/admin/cms/used-vehicles	Admin	Administrative pre-owned stock listing
POST	/api/admin/cms/used-vehicles	Admin	Adds certified pre-owned unit
GET	/api/admin/cms/price-books	Admin	Versioned price book directory
POST	/api/admin/cms/price-books	Admin	Creates new price book draft
PUT	/api/admin/cms/price-books/:id/entries/:vId	Admin	Upserts itemized pricing components
POST	/api/admin/cms/price-books/:id/publish	Admin	Publishes price book with date-overlap check
GET	/api/admin/staff	Admin	Staff user directory
POST	/api/admin/staff	Admin	Provisions new staff user with location assignments
PATCH	/api/admin/staff/:id	Admin	Updates staff user (with last-admin safeguard)
POST	/api/admin/staff/:id/disable	Admin	Deactivates staff user and revokes active sessions